

Fitting the comfort-zone boundary to the study-1 data: plan and methods

2026-08-27. Prepared after the continuous lane-entry forms were settled (docs/lane_entry_note.md) and the desired-speed staging fixed — the two prerequisites named in handover_2026-08-26.md §7. The data pipeline is built (src/comfortzone/czb_data.py), a feasibility fit has been run, and this note records the model, the method, and the decisions that remain. It extends, and where concrete supersedes, section (b) of docs/czb_study1_data_plan.md.

1 How many parameters this actually is

Jonas asked whether we have good methods for fitting "with so many parameters". The answer we would give: the fitting is not high-dimensional in the way that question fears, and the reason is worth stating because it is a design property, not luck.

Everything upstream of the boundary is frozen. The preference function keeps the released parameters (the two structural changes are parameter-free); the desired speed is staged from the clip; the field along each stimulus is therefore a fixed, precomputed curve. Nothing in the perception, preference, or scenario layers is fitted. What is fitted is the *response* layer:

block	parameters	count
population boundary level	mean and sd of c (or of $\log c$)	2
per-driver levels c_i	random effects, shrunk to the population	43 (80 with Button)
response bias b_i	per-driver false-alarm propensity, identified by C1	43 (or 1 group-level)
ordered comfort/dread	one offset Δ between the two nested levels	1–2
accumulator (stage 2)	gain and noise (threshold fixed by c 's scale)	2
paradigm shift δ (stage 3)	one level-or-rate shift for Button	1

Roughly 5–8 population parameters plus ~90–170 random effects. That is a **generalized linear mixed model**, not a black-box search: random effects of this kind are shrunk, not searched over, and every standard tool handles thousands of them. The methods question dissolves once the covariate is precomputed.

Why the likelihood is cheap. The stimulus field is deterministic and shared: all 3 096 Random cut-in trials take one of 18 covariate values (3 criticality levels $\times$ 6 truncation points), computed once. A full likelihood evaluation is vectorized arithmetic over 3 096 rows; a complete MAP fit takes seconds on this machine. Nothing here needs the simulator, GPUs, CMA-ES, or approximate Bayesian computation — methods one reaches for when the likelihood itself requires simulation, which it does not here precisely because the ego never responds in the clips.

2 The model, in three stages

Stage 0 — static threshold, population level (run, as feasibility). Intervention iff the running-max deficit exceeds a threshold, Gaussian variability: $P(\text{intervene}) = \Phi((x - c)/\sigma)$ with $x = \text{deficit_max}$ at clip end. Two parameters against the 18-cell surface: **correlation 0.90, RMSE 0.125** ($c \approx 5\,200$, $\sigma \approx 2\,200$). For a field with zero fitted constants upstream, our reading is that this is strong feasibility evidence. The residual structure is informative: the fit is too flat *in time* within each criticality level (for TTC4 it predicts $0.70 \rightarrow 0.78$ across C3–C6 where the humans go $0.73 \rightarrow 0.92$), which is exactly what a static threshold cannot express and an evidence accumulator can — probability of having crossed grows with exposure time even at a constant covariate. The time gradient the static model misses is the accumulator's signature, visible before fitting one.

Stage 1 — hierarchical, ordered, per driver. The real fit:

- $c_i \sim$ population distribution (fit on log scale; the deficit is positive and the level multiplicative);
- per-driver bias b_i as a lapse/guess floor: $P = b_i + (1 - b_i) \Phi((x - c_i)/\sigma)$, identified by the C1 cells (4 trials per participant per criticality level, so 12 pre-onset-equivalent trials per driver; thin, so b_i gets strong shrinkage or one group level — decision 2 of the handbook ch. 11 list); [decided 2026-08-27 at review gate R.1: hierarchical b_i with

shrinkage is the primary variant — per-driver pre-onset heterogeneity is decisively overdispersed and the group variant leaks pre-onset propensity into the thresholds ($p = -0.622$); group-level is kept as the upper sensitivity bracket; evidence in `replication/czb/out/bias_variant_diagnostics.md`];

- the ordered braking-expectation response (nothing / gentle / hard: 1 180 / 1 287 / 629 trials) as two nested thresholds c_i and $c_i + \Delta$ on the same field — the dread level well identified, the comfort level latent with a wider posterior, per the agreed framing.

Stage 2 — the accumulator, and the Button data as held-out validation. Replace the static threshold with the existing preference-relative accumulator (`src/surprise/`): evidence = the deficit series, crossing = response, giving $P(\text{crossed by } T)$ for the fixed-clip cells and a full press-time density for Button. *[Amended 2026-08-27 at review gate R.1: accumulation — drift and noise both — is gated at manoeuvre onset. The first A.3 run integrated Wiener noise from clip start, through a 15.1 s pre-onset window that is a property of stimulus presentation, not of drivers; the noise-alone running maximum ($sd \approx 3.9$) rivalled the entire post-onset drift, forcing the threshold up and flattening the predicted surface (worklog query A.3.Q1). The gate is observable — onset is the stimulus's first lateral motion, the pipeline's own 0.03 m detection threshold — so this is evidence-gated accumulation, not oracle knowledge; pre-onset responses are carried by the lapse floor exactly as in stage 1. The leaky accumulator was considered and held in reserve: it bounds pre-onset noise equally but also discounts early post-onset evidence, changing the time-integration claim under test, and adds a parameter.]* Fit gain and noise on the Random surface only; predict the Button press-time distributions (2 396 trials, only 0.5% censored — participants essentially always press eventually, so the density is well populated) allowing the single paradigm shift δ , which per the cross-paradigm check should absorb the known excess of early pressing at intermediate criticality (+0.26 at TTC6). Whether δ acts on the level or on the rate is decision 3 of the ch. 11 list; the press-time *shape* distinguishes them, which is why it should be fitted both ways and compared rather than assumed.

Then transfer. Fit on cut-in, predict LTAP, cyclist overtake and truck overtake with every parameter unchanged. That is the falsifiable claim; the elliptical formulation is the tracked fallback. The transfer scenarios need their own field construction first (the cut-in loader mis-assigns roles outside its scenario — section 5).

3 Method recommendation

- **Estimation: MAP plus Laplace, in torch.** Write the joint log posterior as a torch scalar (the covariates are 18 numbers; the random effects are two vectors), optimize with L-BFGS, and take the Laplace covariance at the mode for intervals. *[Corrected 2026-08-27 by card A.1. Taken literally — maximizing the joint posterior over hyperparameters and driver effects together — this recovers the population median and the lapse but inflates the between-driver sd by a factor of about 2.6, with coverage 0/20: the joint mode of a hierarchical posterior is not its marginal mode. Since the deliverable is a population percentile, and a percentile is made of the spread, this would have been a material error. The estimator to use is `replication/czb/fit_recovery.py::fit_marginal`, which integrates the driver effects out by Gauss-Hermite quadrature — one dimension per driver — and applies Laplace only to the four hyperparameters. Evidence in `replication/czb/out/recovery_summary.md`.]* CPU-torch on this machine handles this size trivially. No PyMC or Stan is installed, and nothing here needs them; if full posteriors are ever wanted, a hand-rolled NUTS is not worth it — the honest upgrade is installing numpyro, and the model is small enough that the choice can be deferred.
- **Validate the machinery on synthetic data first:** simulate responses from known c_i , b_i , σ , refit, and confirm recovery — the property-test discipline applied to the fitting code. This also calibrates how much the C1 cells actually constrain b . *[Note added 2026-08-27, review gate R.1: two defects were found in `fit_stage1.py`'s validation code — not in the fits. Its held-out likelihood integrated the hierarchical variant's two random effects on the same quadrature nodes (the diagonal of the 2D integral, asserting perfect rank correlation), so the stored LOPO values for the hier variants (deficit +0.6, a_{req} +14.3) are approximations pending card A.2.v2's regeneration; and its C1 predictive plugged in the hierarchical variant's median lapse (0.017) where the population mean over the lapse distribution is required, understating the pre-onset predictive — corrected numbers in `replication/czb/out/bias_variant_diagnostics.md`. Both functions are fixed in place.]*
- **Uncertainty conventions carry over** from the equivalence work: report intervals, compare conditions as differences with all variance sources resampled, and state what is fixed.
- **Identifiability, stated up front:** the pair (c, σ) is identified only relative to the field's scale, which depends on the structural flags — levels are comparable only between runs with the same flags. Motor latency is not identified by this design at all (fix at a literature 0.2–0.3 s or absorb; ch. 11). The paradigm term δ is a combined modality-plus-order effect and must be named as such.

4 Which axis the boundary lives on: deficit level or allowed deceleration

Two covariates are carried side by side (czb_data):

- `deficit_max` — the level-on-the-field reading: one scalar c per driver on the preference field. Pilot: correlation 0.90 on the surface.
- `a_req_max` — the allowed-deceleration reading: a driver intervenes when the situation demands more braking than they accept. Pilot: correlation 0.73 as currently gated (the lane gate at the C1 boundary frame leaks for TTC8, visible in the cell table), so presently second — but it has a property the deficit lacks: the mild truck cut-ins produce **zero dread-field deficit through most of the response window** (docs/lane_entry_note.md §5) while `a_req` grades smoothly there. If truck participants press at rates a bias term cannot absorb, the deficit-level model fails on trucks and the deceleration axis (equivalently, the comfort field at a lower allowed deceleration) becomes necessary rather than optional.

Our recommendation: fit stage 1 on `deficit_max` as the primary (it is the scalar the CZB claim is about), run the truck cells as the internal check, and keep the deficit-family-over-allowed-deceleration refinement in reserve. This is a place where the data should be allowed to overrule the framing, and the check is cheap.

5 What is built, and what remains before the fit

Built and verified today: trace loading with corrected onset detection, the continuous field, `random_cutin_trials()` (3 096 trials, 43 participants, covariates verified monotone across C2–C6), `button_cutin_trials()` (2 396 trials, press times on the time-since-onset axis, medians ordered by criticality: TTC4 0.36 s → TTC8 1.8 s), and the stage-0 pilot. The response surface reproduces the study's documented table.

Remaining, in order: (1) the synthetic-recovery harness; (2) the stage-1 torch model; (3) truck cut-in trials added to the table (loader handles them; they were left out of the pilot to keep it to the documented 3×6 surface); (4) the accumulator layer; (5) field constructions for the three transfer scenarios — LTAP needs a crossing geometry (the cut-in `x_tar` convention is meaningless there), and the overtake/ truck-lateral traces need role assignment by instructed-vehicle rather than by lateral span, which mis-assigns both (review finding, 2026-08-27).

6 Questions for Jonas before stage 1 runs

(Answered 2026-08-27 — the answers and their consequences are recorded in docs/czb_validation_roadmap.md section 5: both bias variants will be fitted and compared; CZB_2 is treated as the human decision pending a new dataset; motor latency is fixed at 0.25 s with a 0.15/0.35 s sensitivity bracket; the Sequence design stays excluded. The original questions are kept below for the record.)

- 1 Bias per driver with shrinkage, or one group-level bias? (We lean hierarchical with strong shrinkage; the C1 data is thin either way.)
- 2 Is the braking-expectation question (CZB_2) confirmed as *expected required braking* — the reading under which it identifies the dread level — rather than the participant's own intended action? The two designs word it differently and the fitting treats only the Random version as ordered.
- 3 Any objection to fixing motor latency at a literature value rather than absorbing it into c ? It changes nothing for percentiles, only for absolute levels.
- 4 The Sequence design stays excluded (autocorrelation), per decision 8 — confirm this still stands now that it would add three more cut-in stimuli.